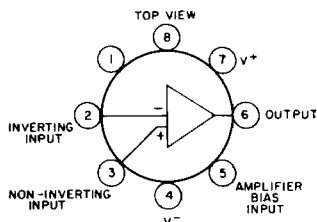


High-Reliability Operational Transconductance Amplifiers Gateable-Gain Blocks

The CA3080 and CA3080A Slash(/) Series types are supplied in the 8-lead TO-5 style package.



NOTE: PIN 8 IS INDICATED BY THE CASE INDEX TAB
92CS-17660

Functional diagram of CA3080
and CA3080A.

TABLE A. POST BURN-IN, FINAL ELECTRICAL AND GROUP A SAMPLING TESTS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS $V^- = -15\text{ V}$, $V^+ = +15\text{ V}$, $I_{ABC} = 0.5\text{ mA}$		LIMITS FOR INDICATED TEMPERATURES ($^{\circ}\text{C}$)						UNITS		
				MINIMUM			MAXIMUM					
				-55	+25	+125	-55	+25	+125			
Input Offset Voltage	V_{IO}		CA3080	—	—	—	6	5	6	mV		
			CA3080A	—	—	—	5	2	5			
Input Offset Current	I_{IO}		CA3080	—	—	—	1.2	0.6	0.7	μA		
			CA3080A	—	—	—	1.2	0.6	0.7			
Input Bias Current	I_I		CA3080	—	—	—	8	5	5	μA		
			CA3080A	—	—	—	8	5	5			
Forward Transcon- ductance	g_m		CA3080	3000	6700	3000	13000	13000	20000	umho		
			CA3080A	4000	7700	4000	9000	12000	18000			
Peak Output Voltage	Positive	$+V_{OM}$	$R_L = \infty$	CA3080	11.6	12	12	—	—	V		
		CA3080A										
	Negative	$-V_{OM}$		CA3080	11.8	12	12	—	—			
		CA3080A										
Peak Output Current	$ I_{OM} $		$R_L = 0$	CA3080	350	350	320	750	650	750	μA	
				CA3080A	350	350	320	750	650	750		
Amplifier Supply Current	I_A			CA3080	0.7	0.8	0.7	1.4	1.2	1.4	mA	
				CA3080A	0.7	0.8	0.7	1.4	1.2	1.4		
Common-Mode Rejection Ratio	C_{MRR}			CA3080	80	80	80	—	—	—	dB	
				CA3080A	80	80	80	—	—	—		
Supply Voltage Rejection Ratio	V_{RR}			CA3080	—	—	—	150	150	150	$\mu\text{V/V}$	
				CA3080A	—	—	—	150	150	150		
Differential Input Current		$I_{ABC} = 10\text{ mA}$, $V_{DIFF} = 4\text{ V}$		CA3080	—	—	—	—	7	—	nA	
				CA3080A	—	—	—	—	5	—		
Magnitude of Leakage Current		$I_{ABC} = 0$, $V_{TP} = 0$		CA3080	—	—	—	—	7	—	nA	
				CA3080A	—	—	—	—	5	—		
			$I_{ABC} = 0$, $V_{TP} = 36$		CA3080	—	—	—	—	7	—	nA
					CA3080A	—	—	—	—	5	—	

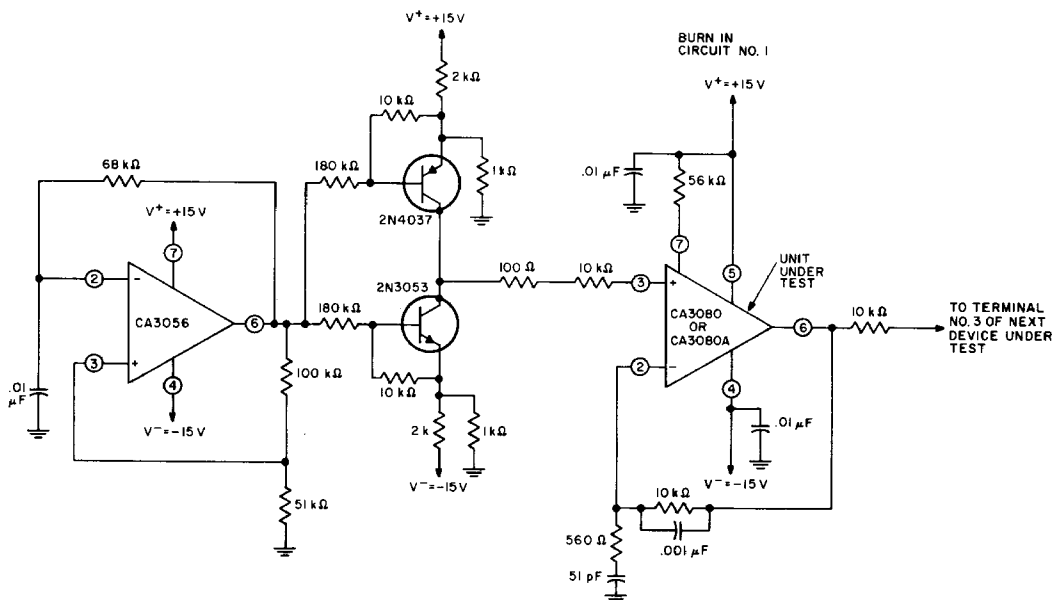
CA3080/..., CA3080A/...

TABLE B. DELTA LIMITS at $T_A = 25^\circ\text{C}$ (/1 only)

CHARACTERISTIC	SYMBOL	TEST CONDITIONS AT $T_A = 25^\circ\text{C}$ $V^+ = +15\text{ V}$, $V^- = -15\text{ V}$	LIMITS	UNITS
			MAX. Δ	
Input Offset Voltage	V_{IO}	CA3080	± 0.2	mV
		CA3080A	± 0.15	
Input Offset Current	I_{IO}	CA3080	± 0.05	μA
		CA3080A	± 0.05	
Input Bias Current	I_I	CA3080	± 0.25	μA
		CA3080A	± 0.25	
Forward Transconductance	g_m	CA3080	± 3000	μmho
		CA3080A	± 3000	

TABLE C. GROUPS C AND D END-POINT TESTS at $T_A = 25^\circ\text{C}$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS $V^+ = +15\text{ V}$, $V^- = -15\text{ V}$	LIMITS		UNITS
			MIN.	MAX.	
Input Offset Voltage	V_{IO}	CA3080	—	6.5	mV
		CA3080A	—	3.5	
Input Offset Current	I_{IO}	CA3080	—	1.2	μA
		CA3080A	—	1.2	
Input Bias Current	I_I	CA3080	—	10	μA
		CA3080A	—	10	
Forward Transconductance to Terminal No. 1	g_m	CA3080	6500	14000	μmho
		CA3080A	7000	13000	
Peak Output Current	$ I_{OM} $	CA3080	300	700	μA
		CA3080A	300	700	
Peak Output Voltage	$+V_{OM}$	CA3080	11	—	V
	$-V_{OM}$	CA3080A	11	—	
		CA3080	—11	—	
		CA3080A	—11	—	



Burn-in and operating life test circuit.

92CM-22842